

WASP-1b

R-1517

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_140908 · created 2026-08-01T16:40:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456908.739 +/- 0.0061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.041
Transit depth (Rp/Rs)^2	1.94 +/- 1.15 [%]
Orbital Inclination (inc)	86.79 +/- 1.46
Airmass coefficient 1 (a1)	4.25 +/- 20.92
Airmass coefficient 2 (a2)	2.999 +/- 0.069
Scatter in the residuals of the lightcurve fit is	50593.14 %
Best Comparison Star	#2 - [617, 490]
Optimal Method	PSF photometry
Transit Duration (day)	0.1523 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.139 ± 0.041 vs 0.1036 (deviation 0.06σ)
Duration fit vs published	0.1523 d vs 0.1567 d (deviation 0.03σ)
Mid-time O–C	11.1 min
Depth SNR	0.2
Residual scatter	50593.14 %

Light curve

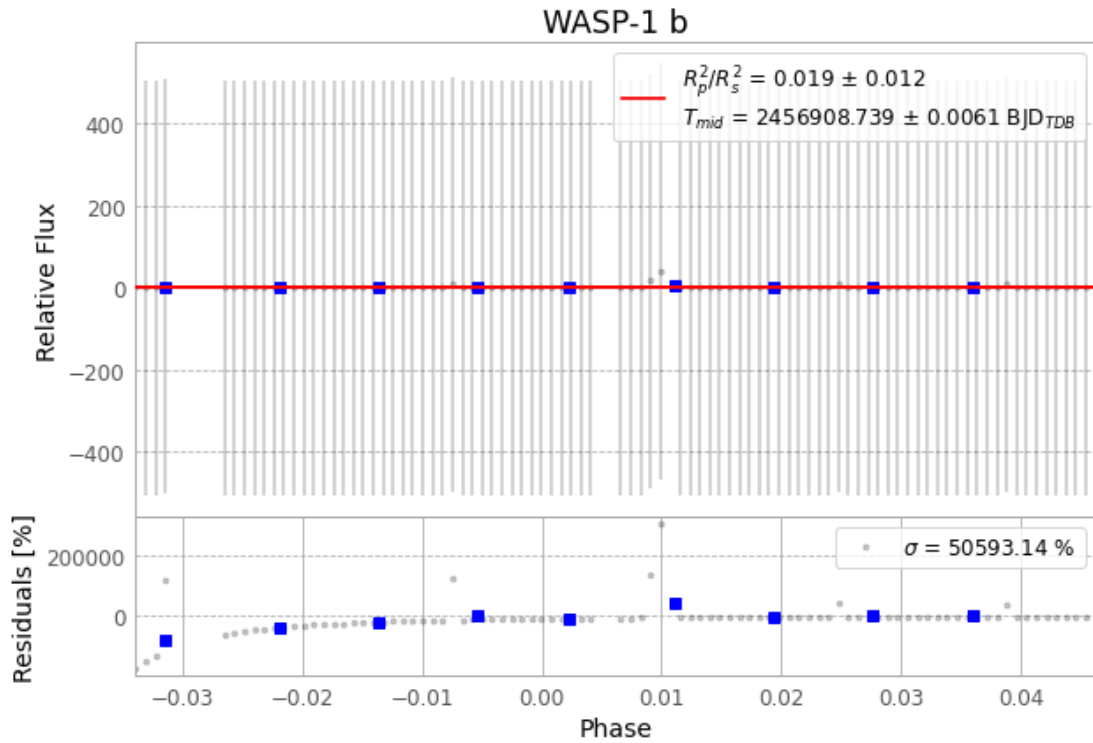


Figure 1 — FinalLightCurve_WASP-1 b_2014-09-07.png (SHA-256 d6adba4b43239340...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [635, 429], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8ffabe281e485c4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

98 FITS frames (64 MB), WASP-1140908033312.FITS → WASP-1140908082412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-140908.log (44a614f622b9...), FinalLightCurve_WASP-1 b_2014-09-07.png (d6adba4b4323...), FinalLightCurve_WASP-1 b_2014-09-07.csv (538100ce6d0e...)

Compute resources

Wall time 475.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 16:40Z.